Graph-e-thon 3.0 Submission

RAKSHA

Modern SOS Safety Platform — One Missed Call. Zero Cost. No Internet.

Team Leader Name: Nishkarsh Jain

Team Member(s): Arnav Sharma, Kush Bhutani, Kanika

College Name: Maharaja Surajmal Institute, New Delhi, India

Tracks: Blockchain for impact, Smart infrastructure, IoT for sustainability, Defence & National Security, Creative tech solutions.

SDG 9: Open Innovation

Problem Statement

Problem Statement

India records 4,48,211 crimes against women/children annually (NCRB 2023). In emergencies, women must unlock a phone, open an app, tap through screens impossible under duress, unavailable on a ₹500 feature phone, and useless without internet.

4,48,211	48%	7-15 min	73%
Crimes against women registered in India (NCRB 2023)	Women/Children feel unsafe in public spaces (India)	Average police response time, urban India	Emergencies occur without smartphone / internet access

Detailed Problem Description

1. **Requires Smartphone & Internet**
   Existing SOS apps like Nirbhaya, SafetiPin, and 112 require smartphones and internet, making them unusable in emergencies and inaccessible on feature phones. Over **60% of rural women rely on basic phones**, and there are **no simple SOS solutions designed for children in panic** situations.

2. **No Tamper-Proof Evidence**
   Evidence submitted digitally is easily manipulated. Indian courts need verified, unchallengeable digital evidence. No existing free platform provides a blockchain-certified FIR package.

3. **High Trigger Cost & Complexity**
   Dialing 112 requires full call connection charges. App-based SOS requires unlocking, opening the app, and tapping buttons — impossible for a woman restrained or under duress.

4. **No NGO Coordination Layer**
   Emergency alerts go only to police. Verified NGO partners — Shakti Foundation, Safe Harbour Trust — with faster on-ground response are never looped in automatically.

Proposed Solution

Solution Statement

RAKSHA enables women and children to trigger an SOS through a simple missed call, requiring no internet and working on any phone (be it a ₹500 feature button phone). It instantly alerts 2 emergency contacts and NGOs while securely recording the incident and generating a legally usable FIR report.

How It Works — SOS Flow

1 → 2 → 3 → 4 → 5 → 6

1. **Missed Call** — User dials Exotel virtual number → drops
2. **Webhook Trigger** — HMAC-verified POST to /api/sos
3. **Incident Created** — PostgreSQL record with location + timestamp
4. **Alerts Dispatched** — SMS + Email + Push + NGO in parallel
5. **Evidence Hashed** — SHA-256 + S3 upload + blockchain mined
6. **FIR Compiled** — Court package with Ethereum anchor hash

Core Features

- **Missed Call SOS:** Works on any phone, zero data, zero cost. Exotel + KooKoo IVR redundant triggers.
- **Multi-Channel Alerts:** Simultaneous SMS (MSG91/Textlocal fallback), Gmail SMTP, Firebase push, NGO email.
- **Aadhaar Verification:** UIDAI-integrated user identity. Prevents fake accounts. Fernet AES encryption at rest.
- **Blockchain Evidence:** SHA-256 PoW chain (difficulty=4). Each file hashed, mined, optionally anchored to Ethereum Sepolia.
- **Court-Ready FIR:** Auto-compiled FIR with blockchain hash, Ethereum tx, GPS location. Admissible under IT Act 2000 §65B.
- **Safety Heatmap:** Folium + scikit-learn anonymized incident density maps for urban planning and policing.

TECHNICAL APPROACH

SYSTEM ARCHITECTURE

Designed for low-connectivity, high-stress scenarios using telecom-triggered, event-driven architecture.

1. **Trigger Layer**
   Exotel Missed Call · KooKoo IVR · App SOS Button
2. **Webhook & Validate**
   HMAC SHA-256 · Flask /api/sos · Rate-limited 60 req/s
3. **Service Layer**
   Alert · Evidence · Blockchain · FIR · Abuse Detection
4. **Data & Storage**
   PostgreSQL 15 · Redis 7 · Azure Blob · Celery 5.4 async queue
5. **Infra & Proxy**
   Nginx SSL/HTTP2 · Gunicorn×4 · 5-container Docker

RAKSHA VS EXISTING SOLUTIONS

Feature	112 / Dial	SafetiPin	RAKSHA
Works without internet	✗	✗	✓
Works on feature phone	✗	✗	✓
Zero cost to trigger	✗	✗	✓
Blockchain evidence	✗	✗	✓
Court-ready FIR	✗	✗	✓
Aadhaar verified users (Future)	✗	✗	✓
NGO auto-alert	✗	Partial	✓
Live heatmap analytics	✗	Partial	✓
Evidence hash chain	✗	✗	✓

FLASK • POSTGRESQL • REDIS • DOCKER • AZURE • TWILIO/EXOTEL • MSG91 • FIREBASE • CELERY • SHA-256 BLOCKCHAIN

BLOCKCHAIN EVIDENCE CHAIN

PROOF-OF-WORK MINING PROCESS

1. User uploads audio/image evidence file
2. SHA-256 hash computed from raw bytes (not metadata)
3. Block assembled: evidence_hash + prev_hash + timestamp
4. **Nonce search: find n where SHA-256(block+n) starts with "000"**
5. Valid block appended — chain integrity maintained
6. Optional: block hash anchored to Ethereum Sepolia testnet

COURT-ADMISSIBLE FIR PACKAGE (IT ACT 2000 | $65B)

- Incident timestamp + GPS / cell-tower location
- Blockchain block index, block hash (starts with 0000)
- Ethereum Sepolia tx hash for independent public verification
- SHA-256 checksums of all uploaded evidence files
- Full audit log: webhook receipt → alert dispatch → chain mine
- Exported via /api/fir/compile/{incident_id} — PDF-ready

BUSINESS MODEL CUSTOMER TYPES

NGO SaaS	NGO funding for verified partner dashboard, incident analytics, and heatmap API access. (Women/Child Support Program)
Govt Contracts	MoU with state police / Nirbhaya Cell for bulk SOS infrastructure. Per-incident licensing cost.
Legal Aid Counsil	FIR package generation API for lawyers and legal NGOs. Pay-per FIR package generation
Data Licensing	Anonymized and aggregated safety insights to government bodies and urban planners. Identify high-risk zones (crime hotspots)

BUSINESS MODEL ROADMAP

REVENUE MODEL (POST-LAUNCH)

RAKSHA follows a scalable subscription model for institutions.
₹10–₹20 per active user/month

What it Includes:	• Unlimited SOS triggers • Emergency contact + NGO alerts • Dashboard & incident tracking • FIR generation system • Basic safety analytics
Customers:	• Government bodies (primary revenue) • NGOs & safety organizations • Legal Council Aid, Smart City Urban Planners
System Execution Flow :	• Simple and consistent pricing • Scales with number of users • Low cost, high-margin infrastructure model
Future Expansion:	• Advanced analytics for government planning • Integration with Smart City safety systems

GO-TO MARKET STRATEGY

Phase 1

- MVP live on Azure + Docker, 91 tests passing
- Partner with 3–5 NGOs in one city
- Pilot deployment with 500–1000 users
- Free usage to validate response system

Phase 2

- WhatsApp Business API alerts
- Real-time location WebSocket
- Approach State Police with pilot data
- Apply under Nirbhaya Fund / Smart City programs

Phase 3

- Expand to multiple states
- Integrate with 112 system
- Scale NGO network nationally
- Govt MoU negotiations

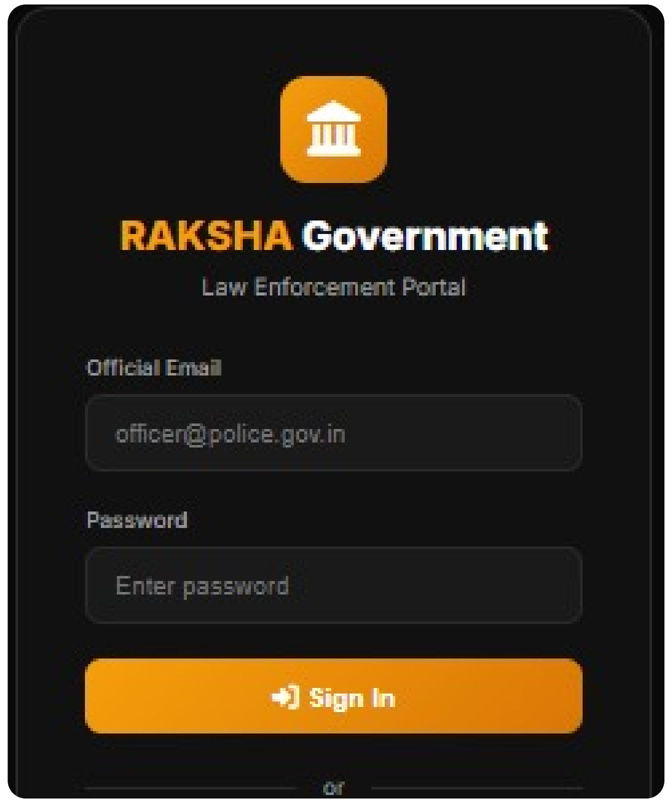
IMPACT AND BENEFITS

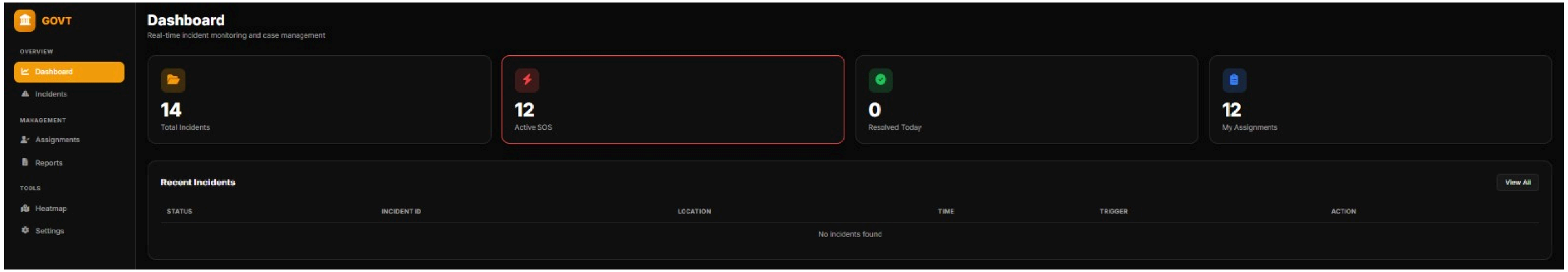
IMPACT METRIC	DETAILS
Social Impact	• Works on any phone → accessible to rural women & children, even in a situation of duress using quick acess buttons • Faster SOS response via NGO and Police Stations + Emergency Contact alerts • Improves reporting, accountablility and data management
Economic Impact	Zero infrastructure cost for the end user — a missed call is free on any network. Celery async dispatch and Docker containerization scale to millions of users with no linear cost increase. Reduces police overhead via intelligent abuse-detection velocity analysis. Reduces legal cost via digital FIR compilation.
Environmental Impact	100% paperless evidence and FIR workflow — eliminates manual court documentation. Cloud-first Docker architecture reduces hardware footprint. No dedicated hardware required; runs on existing ₹500 GSM phones. Serverless-ready design minimizes idle resource consumption.
Target Audience	Primary: Children /Women aged 9–60 across India (urban and rural). Secondary: Verified NGO partners — women's/child shelters, legal aid, crisis centers. Tertiary: Law enforcement and courts (FIR packages). Quaternary: Urban planners and policy makers (anonymized safety heatmap data).

PROTOTYPE

Live Running Application — Flask v1.0 · Zero Cost · Zero Internet

① GOVERMENT DASHBOARD

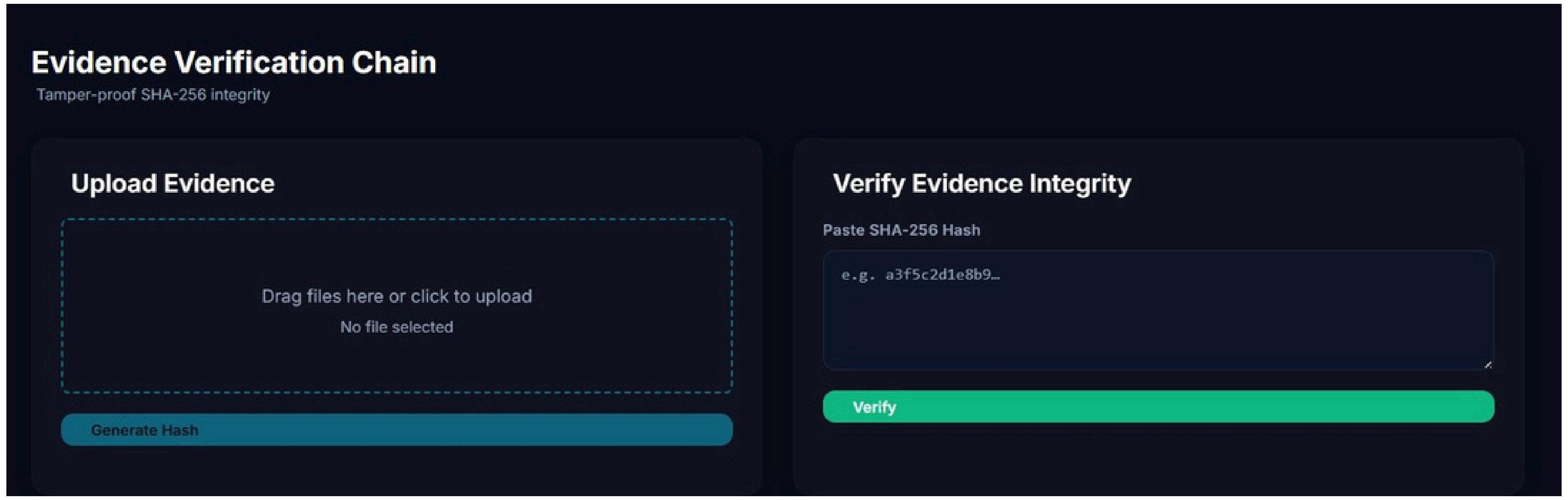




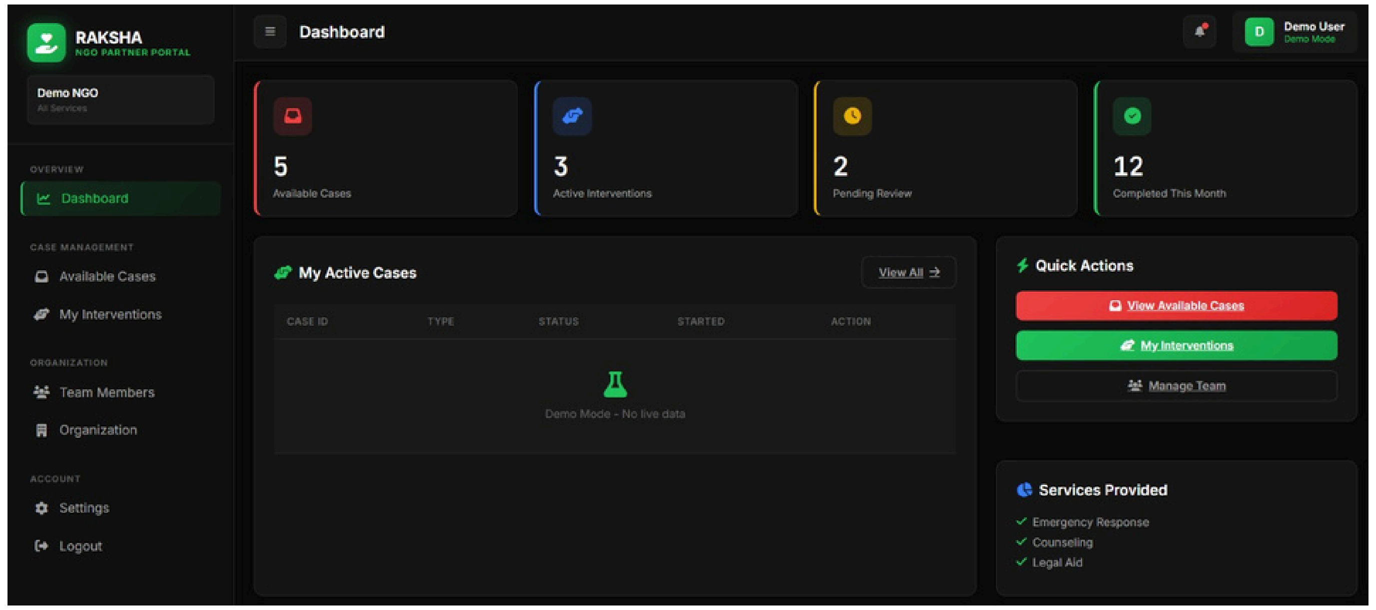
KEY FEATURES VISIBLE:

- Live Active SOS counter
- Blockchain Evidence Ledger panel
- Chain Valid ✓ status
- Real-time incident table with LIVE badge
- Avg response time < 10 seconds
- Sidebar: Dashboard · Live Map · Evidence · Partners · Evidence Chain

② EVIDENCE VERIFICATION CHAIN



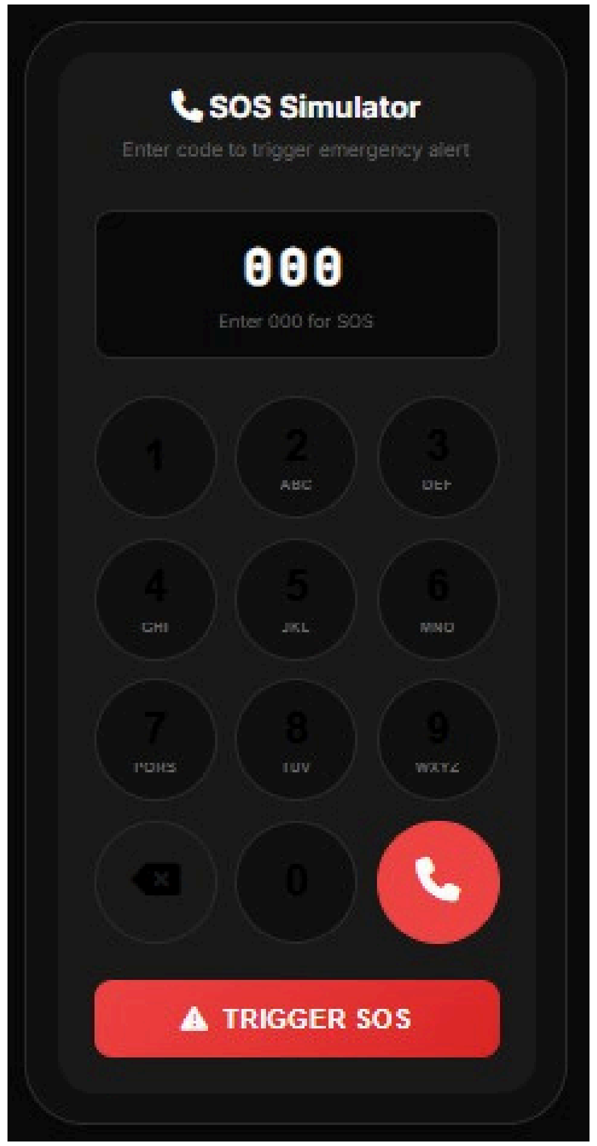
③ NGO PARTNER MANAGEMENT

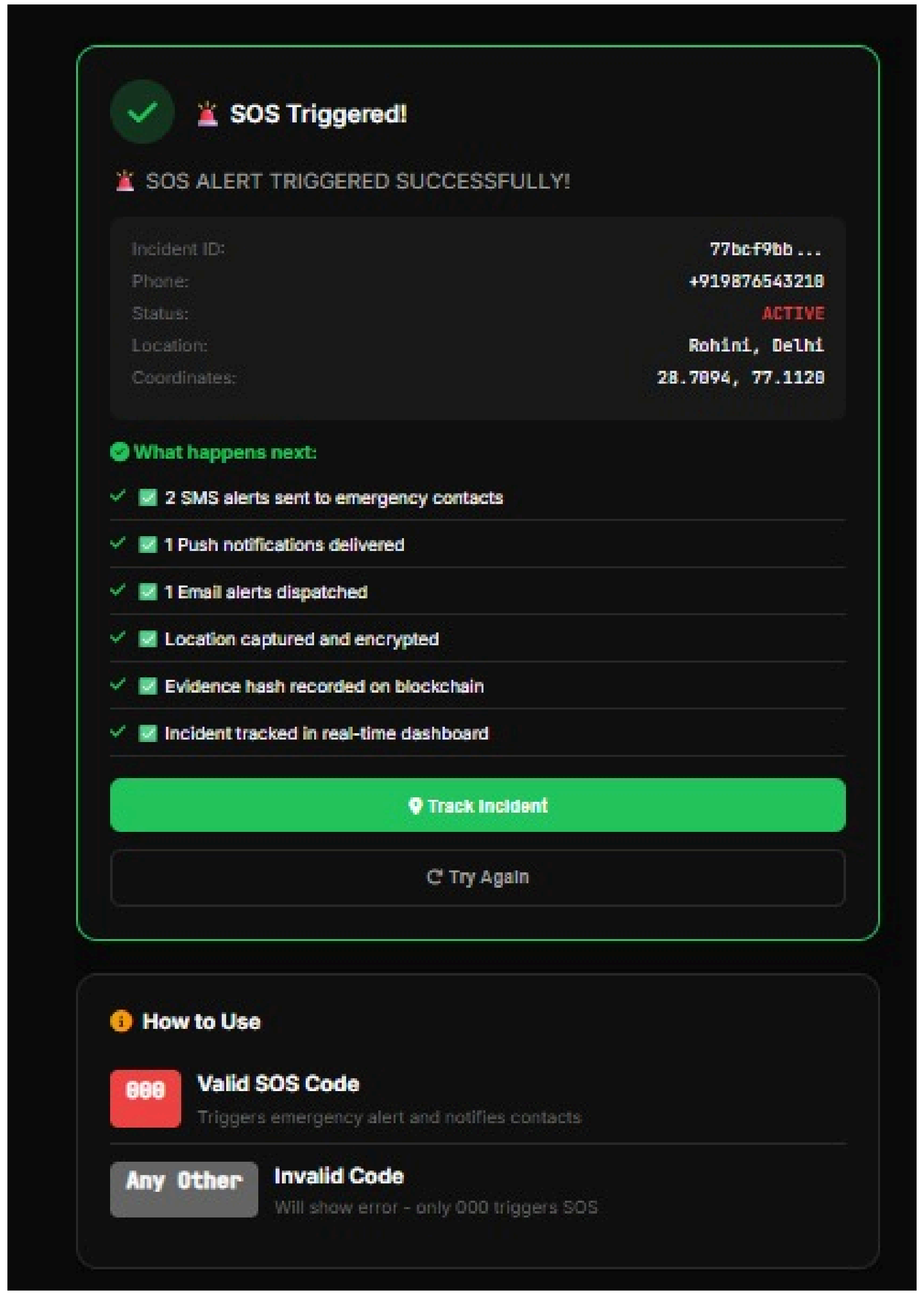


PROTOTYPE DEMO VIDEO

PROTOTYPE & REFERENCES

SOS TRIGGER USER END





WHAT'S HAPPENING:

- User is on a dial-pad style interface
- They enter a code (000) to simulate SOS
- This mimics the real-world missed call trigger

CONFIRMATION:

- "SOS Triggered Successfully"
- Incident ID generated
- Phone + location captured

REPOSITORY & PROJECT STATS

Repo Link: https://github.com/restricted2008/Raksha

41 Git commits	**94** Tests passed
33 Python packages	**50** REST API routes

RESEARCH & REFERENCES

- **NCRB 2023 Report :** 4.4+ lakh crimes against women/children → validates problem scale
- **TRAI / GSMA Reports :** High feature phone usage in rural India → supports no-app approach
- **IT Act 2000 — Section 65B :** Digital evidence admissible in court → supports FIR system
- **SHA-256 Standard (NIST) :** Ensures data integrity and tamper-proof evidence
- **Telecom & SMS APIs (Exotel, MSG91) :** Proven infrastructure for missed call and alert systems